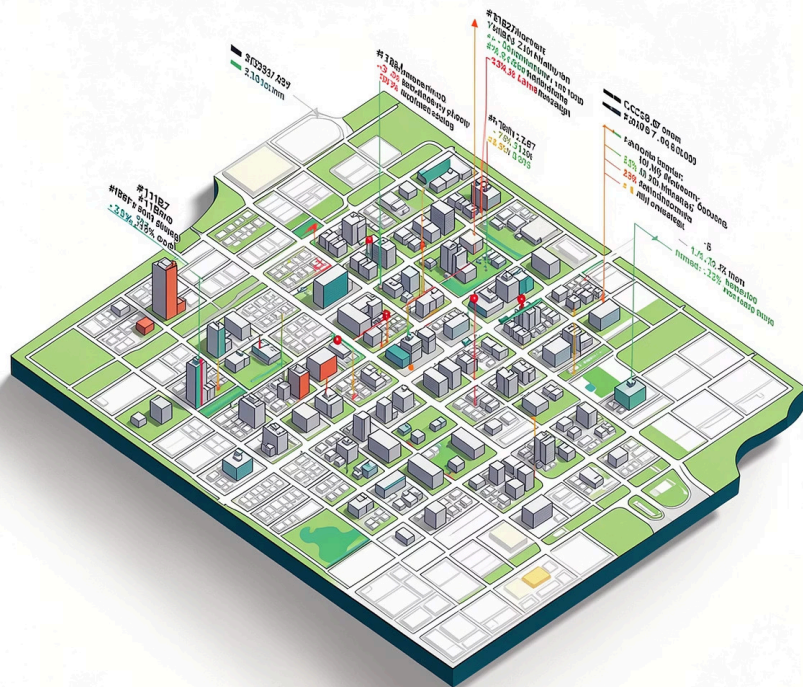




GeoSpatial Intelligence Dashboard

A full-stack GIS platform integrating spatial analytics, decision intelligence, and real-time communication — built for modern, data-driven geographic decision-making.

NEXT.JS

LEAFLET

POSTGIS

MONGODB

Project Overview

This platform delivers an interactive environment where users explore geographic data and make informed, location-aware decisions. It unifies three core capabilities into a single cohesive system.

GIS Mapping

Dynamic, interactive spatial visualization powered by real-time data layers

Analytics

Location-based scoring, KPIs, and data-driven insights surfaced instantly

Collaboration

Integrated chat and video calling for seamless team decision-making



Problem Statement



The Gap in Traditional Systems

Conventional decision-support platforms fall short in two critical areas:

- i Users require a system that delivers dynamic, location-based decision support in real time.

No Geographic Visualization

Static reports and tabular data fail to convey spatial relationships or location-specific context.

Lack of Real-Time Insights

Delayed data pipelines leave decision-makers without the live intelligence needed to act with confidence.

Solution Architecture

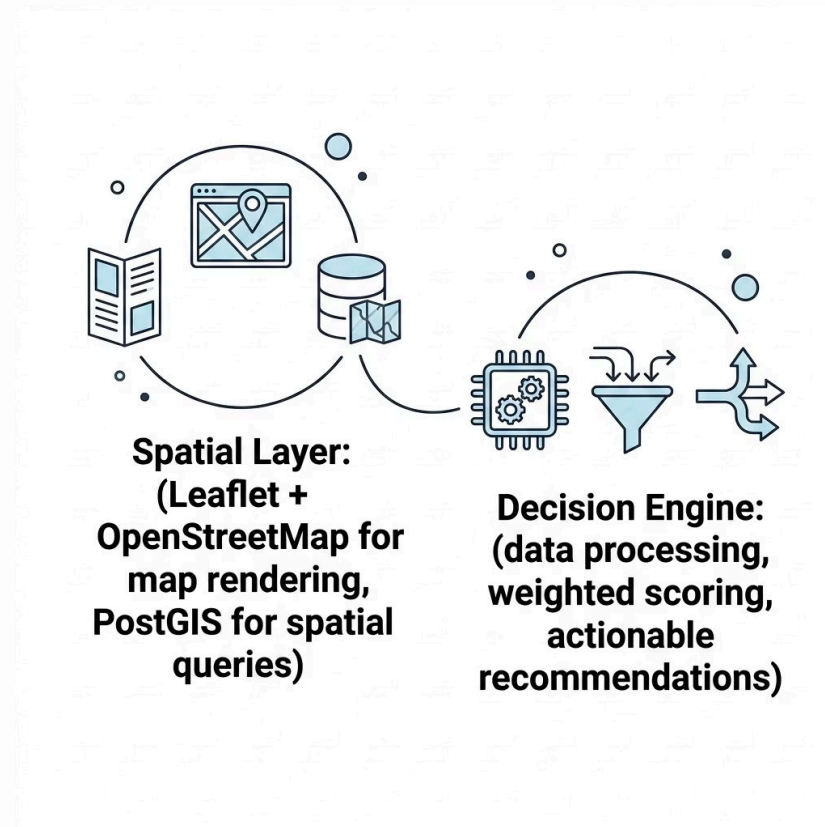
The platform combines best-in-class open-source mapping with a purpose-built decision engine to transform raw spatial data into actionable intelligence.

Spatial Layer

Leaflet integrated with OpenStreetMap provides fluid, interactive map rendering. PostGIS powers high-performance spatial queries — enabling precise geospatial lookups at scale.

Decision Engine

A server-side processing layer continuously evaluates incoming data, applies weighted scoring models, and surfaces clear, actionable recommendations for each geographic area.



Technology Stack



Frontend

Next.js with Tailwind CSS — server-side rendering, responsive UI, and rapid component development



Backend

Next.js API routes providing a unified, scalable server architecture with minimal overhead



Database

PostgreSQL with PostGIS for spatial data; MongoDB for persistent chat and session storage



Real-Time

Socket.io for live data streaming; WebRTC enabling peer-to-peer video communication

The interactive map engine supports fluid zoom, pan, and real-time layer updates – ensuring users always interact with the most current spatial data available.

As users navigate the map, the system dynamically computes the visible bounding box and issues targeted PostGIS queries – fetching only the data relevant to the current view. This approach minimizes payload size and maximizes rendering performance.

- ✔ Result: A responsive, high-performance map experience that scales with data volume.

Decision Intelligence

The decision engine performs multi-factor spatial analysis to evaluate any selected location, delivering a composite score and a clear, evidence-based recommendation.



Population Density

Demographic reach and market potential



Crime Index

Safety profile derived from incident data



Infrastructure

Transport access, utilities, and connectivity



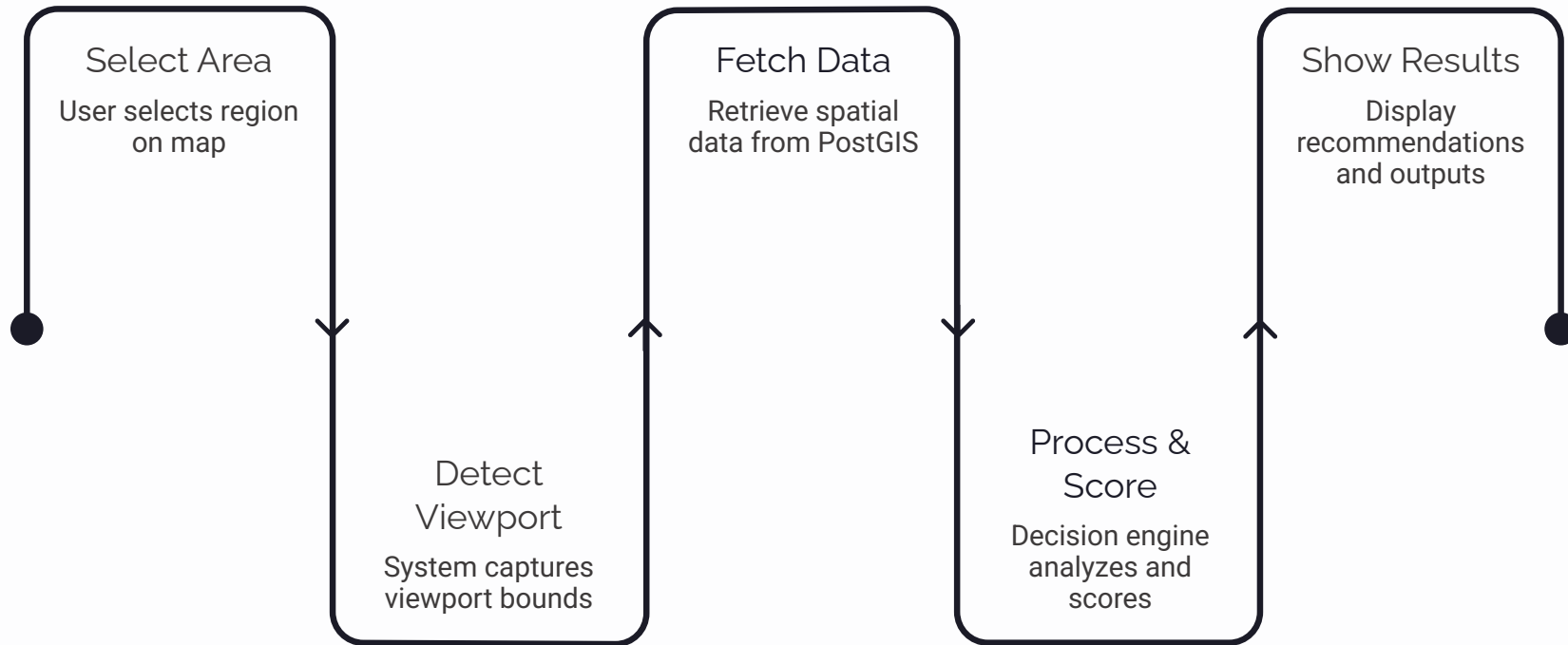
Competition

Proximity and density of competing entities

 Each factor is weighted and aggregated into a unified location score with an accompanying recommendation.

User Workflow

The end-to-end workflow is designed for speed and clarity – from initial area selection to actionable output, every step is automated and transparent.



The pipeline executes in near real-time, ensuring decision outputs are available within seconds of user interaction – enabling rapid, iterative analysis across multiple locations.

UI & Platform Features



Split-View Dashboard

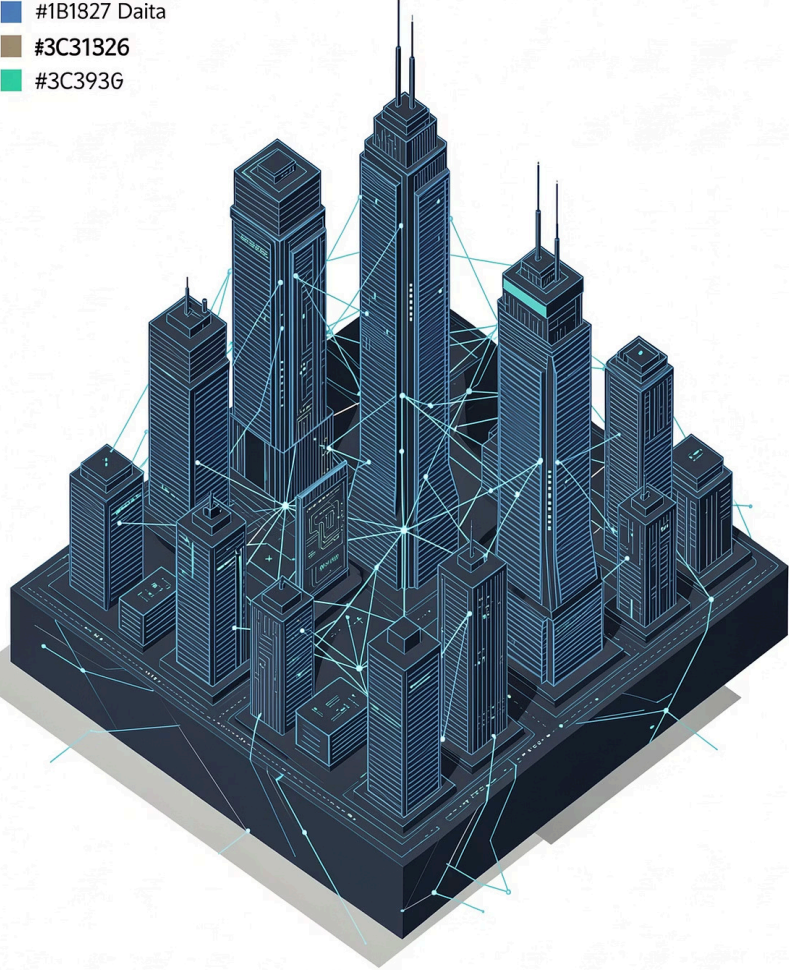
The primary interface presents a synchronized split view — the interactive map occupies one panel while the analytics pane surfaces contextual KPIs, trend charts, and scoring data in real time.

→ Charts & KPI Cards
Dynamic visualizations update as the user explores different locations

→ Integrated Chat
Persistent team messaging backed by MongoDB for context-aware collaboration

→ Video Calling
WebRTC-powered peer-to-peer video directly within the dashboard

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Future Work & Conclusion

Roadmap

Expanded Data Integration

Automated pipelines to ingest third-party datasets — real estate, economic, and environmental sources

AI-Powered Insights

Machine learning models to predict location suitability and surface proactive recommendations

Conclusion

The GeoSpatial Intelligence Dashboard represents a scalable, production-grade platform that bridges the gap between raw geographic data and confident, real-world decision-making.

By unifying spatial analysis, decision intelligence, and real-time collaboration, it empowers organizations to act with precision — wherever decisions need to be made.